Benchmarking Long-Sequence Attention and State-Space Mechanisms in Deep Unfolding Networks for Limited-Angle CT Reconstruction

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Abstract. Limited-angle computed tomography (LA-CT) is an ill-posed inverse problem that suffers from severe missing-wedge artifacts due to incomplete angular scanning geometries. Deep unfolding networks, which unroll iterative optimization algorithms into cascading neural stages, offer physics-guided reconstruction by interleaving data fidelity operations with learned regularization priors. However, conventional CNN-based regularizers suffer from restricted receptive fields that cannot resolve non-local geometric streak trajectories across the angular void, while standard Vision Transformers incur intractable $\mathcal{O}(N^2)$ quadratic computational complexity on high-resolution CT grids. In this work, we conduct a systematic empirical benchmark evaluating three emerging linear and sub-quadratic long-sequence modeling architectures integrated as learnable regularizers within the LEARN unrolling framework: (1) sliding-window attention with global tokens (**Longformer**), (2) multi-scale exponentially dilated attention (**LongNet**), and (3) selective state-space sequential models (**Mamba**), benchmarked comprehensively against both the canonical CNN-based unrolling baseline (**LEARN**) and a dual-domain transformer architecture (**DuDoTrans**). Comprehensive experiments on the clinical Mayo Clinic Low-Dose CT (LDCT) benchmark across moderate (120°) and severe (90°) limited-angle regimes reveal critical architectural behaviors: (i) Under moderate 120° scanning, **LEARN_Longformer** attains the highest reconstruction PSNR (33.10 dB) and lowest RMSE (0.0224), improving over canonical CNN LEARN (32.29 dB, +0.81 dB) and dual-domain DuDoTrans (25.15 dB, +7.95 dB), demonstrating that localized windowing paired with sparse global anchors effectively suppresses non-local streaks, while CNN LEARN maintains higher structural similarity (0.9383 vs. 0.9237 SSIM) due to local convolutional smoothing priors; (ii) **LEARN_LongNet** delivers robust multi-scale artifact suppression (31.62 dB PSNR, 0.8991 SSIM); (iii) Under extreme 90° truncation (missing 180° Fourier angular wedge, corresponding to a 50% frequency null space under Hermitian symmetry), unconstrained 16,384-token sequence regularizers stall around 19.2 dB PSNR without physical gradient guidance, whereas canonical CNN LEARN preserves conservative stability (28.03 dB PSNR, 0.8793 SSIM) and dual-domain inpainting in **DuDoTrans** exhibits resilience (21.81 dB PSNR) by interpolating projection trajectories prior to backprojection; and (iv) **LEARN_Mamba** achieves 26.32 dB PSNR at 120° but undergoes acute performance degra-

dation under severe 90° truncation (18.76 dB PSNR, 0.4292 SSIM), uncovering the structural vulnerability of 1D causal scanning when encountering 2D anisotropic frequency voids. Our findings provide concrete empirical guidelines and architectural trade-off insights for modern deep unrolling models in tomographic imaging.

Keywords: Limited-Angle CT · Deep Unfolding · Long-Sequence Modeling · Transformer · Longformer · LongNet · Mamba · DuDoTrans · Empirical Benchmark.

1 Introduction

X-ray Computed Tomography (CT) is an indispensable non-invasive diagnostic modality in modern healthcare [19, 10]. However, in numerous specialized clinical and interventional scenarios—such as C-arm intraoperative imaging, dental tomosynthesis, and image-guided radiotherapy—physical mechanical clearance or scanning time constraints restrict the gantry's angular coverage to a limited angular range (e.g., 120° or 90° instead of the full 180° or 360°) [5, 17]. This condition, known as Limited-Angle CT (LA-CT), represents a severely ill-posed inverse problem. According to the Fourier Slice Theorem [10], incomplete angular sampling creates a non-sampled "missing wedge" in Fourier frequency space. Direct analytical inversion using Filtered Backprojection (FBP) yields heavy geometric streak artifacts, severe edge distortion, and substantial loss of diagnostic contrast perpendicular to the missing projection rays [5, 17]. Under a 90° scanning arc, the missing frequency void corresponds strictly to a 180° Fourier angular wedge (50% missing frequency area in the 2D Fourier plane governed by Hermitian symmetry).

To address this challenge, classical iterative reconstruction algorithms (such as Total Variation minimization [17]) were proposed, but they often incur prohibitive computational overhead and hand-crafted hyperparameter tuning. Recently, model-based deep learning and deep unfolding networks (such as LEARN [5], Learned Primal-Dual [1], MoDL [2], ISTA-Net [23], and RegFormer [21]) have emerged as an effective reconstruction framework. Deep unfolding cascades iterative gradient descent updates across $K = 14$ stages:

$$x^{(k+1)} = x^{(k)} - \left[\alpha^{(k)} A^T (Ax^{(k)} - y) + \mathcal{R}_{\theta^{(k)}}(x^{(k)})\right], \tag{1}$$

where A denotes the forward Radon projection operator, y represents the acquired sinogram, A^T is the adjoint backprojection operator, $\alpha^{(k)}$ is a learnable physical step size, and $\mathcal{R}_{\theta^{(k)}}$ is a learned neural regularizer. In numerical tomographic implementations, to accelerate convergence and condition the spectrum of the normal operator $A^T A$, preconditioned gradient descent with $A^{\dagger}_{\text{FBP}} = PA^T$ (where P is a ramp/Ram-Lak filtering operator) is typically adopted, as originally formulated in LEARN [5]. By embedding the exact physical projection geometry into the data consistency term, deep unfolding guarantees physical

fidelity while harnessing the expressive power of deep neural networks. Dual-domain extensions like DuDoNet [11] and DuDoTrans [20] further couple sinogram inpainting with image-domain restoration.

Despite its mathematical elegance, the reconstruction quality of deep unfolding is fundamentally governed by the representation capacity of the regularizer $\mathcal{R}_\theta$. In transmission tomography, measurement acquisition is inherently non-local: each sinogram detector value represents a continuous line integral along an X-ray trajectory traversing the entire patient anatomy [10]. Consequently, incomplete angular sampling does not merely cause localized pixel blurring; rather, missing-wedge frequency voids propagate coherent, high-amplitude streak artifacts globally across distant anatomical regions along the boundary projection tangents [4]. Existing deep unfolding architectures predominantly utilize standard Convolutional Neural Networks (CNNs) as regularizers [5, 16]. Because convolutional kernels operate over strictly localized receptive fields (typically 3×3 or 5×5), CNNs face inherent limits when tracing and suppressing these spatially extensive, cross-field-of-view streak trajectories.

To capture non-local context, Vision Transformers (ViTs) [8, 18] and hierarchical attention architectures [12] employ self-attention mechanisms. However, conventional ViTs partition images into coarse patches (typically 16×16 or 8×8 pixels). In diagnostic medical CT, delicate pathological anomalies, microcalcifications, and high-contrast trabecular bone margins are often only a few pixels wide. Coarse patch tokenization imposes an artificial locality bias, smoothing out critical diagnostic micro-structures. Conversely, resolving fine anatomical details requires fine-grained tokenization—ideally using $p \times p$ tokens with $p = 2$ pixels [4]. On a standard 256×256 reconstruction grid, this fine granularity expands the sequence length to $N = (256/2) \times (256/2) = 16,384$ tokens. Standard full self-attention incurs quadratic computational and memory complexity $\mathcal{O}(N^2)$, requiring the computation of an attention matrix of size $16,384 \times 16,384 \approx 2.68 \times 10^8$ elements per head at every unrolling stage. Across a 14-stage unrolled cascade, this creates an intractable memory bottleneck that prevents end-to-end training on modern GPUs.

To bridge this fundamental trade-off, recent breakthroughs in sequence modeling have proposed sub-quadratic and linear-complexity architectures. These include:

1. **Longformer** [3]: Employs sliding-window localized attention with sparse global memory tokens, achieving $\mathcal{O}(N \cdot w)$ linear complexity.
2. **LongNet** [7]: Introduces dilated multi-scale attention patterns with exponentially expanding receptive fields, handling sequence lengths up to one billion tokens.
3. **Mamba** [9, 6]: Leverages selective structured state-space models (SSMs) to accomplish linear-time sequence modeling with recurrent fast scanning, recently explored in vision (Vim [24]) and medical imaging (U-Mamba [13]).

While these architectures have demonstrated strong results in natural language and general vision tasks, their relative strengths, inductive biases, and

failure modes when embedded within the iterative physical unrolling pipeline for ill-posed CT inversion have not been systematically investigated.

Contributions of this work:

- **Systematic Multi-Architecture Empirical Benchmark:** We establish a rigorous empirical benchmark comparing three modern sub-quadratic sequence models (Longformer, LongNet, and Mamba) against the canonical CNN-based deep unfolding baseline (LEARN [5]) and the representative dual-domain transformer architecture (DuDoTrans [20]) for LA-CT.
- **Multi-Regime Clinical Evaluation:** We evaluate all models on the clinical Mayo Clinic Low-Dose CT dataset across two distinct angular scarcity levels: moderate (120°) and severe (90°), comparing quantitative metrics (PSNR, SSIM, RMSE), parameter efficiency, and runtime latency.
- **Objective Metric Trade-Off and Failure Mode Analysis:** We analyze why **Longformer** achieves peak PSNR performance at 120° (33.10 dB PSNR / 0.0224 RMSE), improving over canonical CNN LEARN (+0.81 dB PSNR) and DuDoTrans (+7.95 dB PSNR) via non-local streak cancellation while CNN LEARN preserves higher SSIM (0.9383 vs. 0.9237); why **DuDoTrans** exhibits higher resilience than image-domain sequence unrolling models under extreme 90° voids (21.81 dB vs. $\approx$ 19.2 dB) via projection-space inpainting; and why **Mamba** collapses under severe missing angles (90°) due to 1D scan anisotropy.

2 Background and Problem Formulation

2.1 Forward Model of Limited-Angle CT

In transmission computed tomography, projection measurement is modeled by the line integral of the tissue attenuation coefficient $x \in \mathbb{R}^N$ along X-ray paths $L(\theta, s)$, parameterized by projection angle $\theta \in [-\Theta_{\text{scan}}/2, \Theta_{\text{scan}}/2]$ and detector channel coordinate s:

$$y(\theta, s) = \mathcal{P}\{x\}(\theta, s) = \int_{L(\theta,s)} x(u, v)\, dl + \epsilon, \tag{2}$$

where $\mathcal{P}$ represents the 2D Radon transform [15] and $\epsilon \sim \mathcal{N}(0, \sigma^2)$ models measurement noise. In discretized matrix-vector notation:

$$y = Ax + \epsilon, \tag{3}$$

where $A \in \mathbb{R}^{M \times N}$ is the forward projection matrix ($M \ll N$), and $x \in \mathbb{R}^N$ is the flattened image vector ($N = H \times W$).

In standard 2D tomographic imaging, exact image reconstruction requires an angular scanning span of at least π for parallel-beam geometries, or $\pi + 2\gamma_m$ (where γ_m denotes the fan half-angle) to satisfy Parker's short-scan sufficiency condition in fan-beam configurations [10]. In limited-angle CT, physical or mechanical clearance constraints restrict the gantry sweep to $\Theta_{\text{scan}} < \pi$ (e.g.,

$\Theta_{\text{scan}} = \frac{2\pi}{3}$ for 120°, or $\Theta_{\text{scan}} = \frac{\pi}{2}$ for 90°). By the continuous 2D Fourier Slice Theorem [10]:

$$\mathcal{F}_{2D}\{x\}(\omega\cos\theta, \omega\sin\theta) = \mathcal{F}_{1D}\{y(\theta,\cdot)\}(\omega). \tag{4}$$

Because real physical attenuation coefficients satisfy Hermitian symmetry in frequency space ($F(-u,-v) = F^*(u,v)$), an acquisition over angular range $[-\Theta_{\text{scan}}/2, +\Theta_{\text{scan}}/2]$ simultaneously determines frequency values in opposite angular sectors. Under a 90° limited-angle scan ($\Theta_{\text{scan}} = \pi/2$), projection rays cover two opposite 90° wedges totaling 180°, leaving strictly an angular void of 180°/360° (50% area of the 2D Fourier frequency domain). In 2D tomography, incomplete coverage of 90° does not represent a 270° missing frequency cone, as each physical X-ray path simultaneously resolves antipodal frequency lines across the origin. The normal operator A^TA is therefore ill-conditioned with an extensive null space $\mathcal{N}(A)$, rendering classical linear inversion like Filtered Backprojection (FBP) plagued by severe directional streak artifacts.

2.2 Deep Unfolding Framework: LEARN

To resolve the missing-wedge ambiguity, image reconstruction is formulated as a regularized optimization objective:

$$\min_x \frac{1}{2}\|Ax - y\|_2^2 + \lambda\Phi(x), \tag{5}$$

where $\frac{1}{2}\|Ax - y\|_2^2$ enforces data consistency and $\Phi(x)$ is a regularizing image prior.

The LEARN framework [5] unrolls the gradient descent minimization of Eq. (5) into a finite cascade of $K = 14$ unrolling stages. Initialized from the analytical filtered backprojection reconstruction $x^{(0)} = x_{\text{FBP}} = A_{\text{FBP}}^{\dagger}y$, the image estimate at stage $k \in \{0, 1, \ldots, K-1\}$ is updated via:

$$x^{(k+1)} = x^{(k)} - \left[\alpha^{(k)}A_{\text{FBP}}^{\dagger}(Ax^{(k)} - y) + \mathcal{R}_\theta\left(x^{(k)}\right)\right], \tag{6}$$

where $\alpha^{(k)} \in \mathbb{R}^+$ is a learnable physical step size, $A_{\text{FBP}}^{\dagger} = PA^T$ denotes the preconditioned backprojection operator utilizing a Ram-Lak filter P to accelerate spectral convergence, and $\mathcal{R}_\theta$ is the learned regularization module with parameter weights θ shared across all $K = 14$ unrolling stages ($\theta^{(0)} = \cdots = \theta^{(K-1)} \equiv \theta$) to preserve stationary mapping and parameter efficiency. The network is trained end-to-end to minimize the Mean Squared Error (MSE) reconstruction loss across paired training samples $\{(y_i, x_i^*)\}_{i=1}^S$:

$$\mathcal{L}(\theta, \{\alpha^{(k)}\}) = \frac{1}{S}\sum_{i=1}^{S}\left\|x_i^{(K)} - x_i^*\right\|_2^2. \tag{7}$$

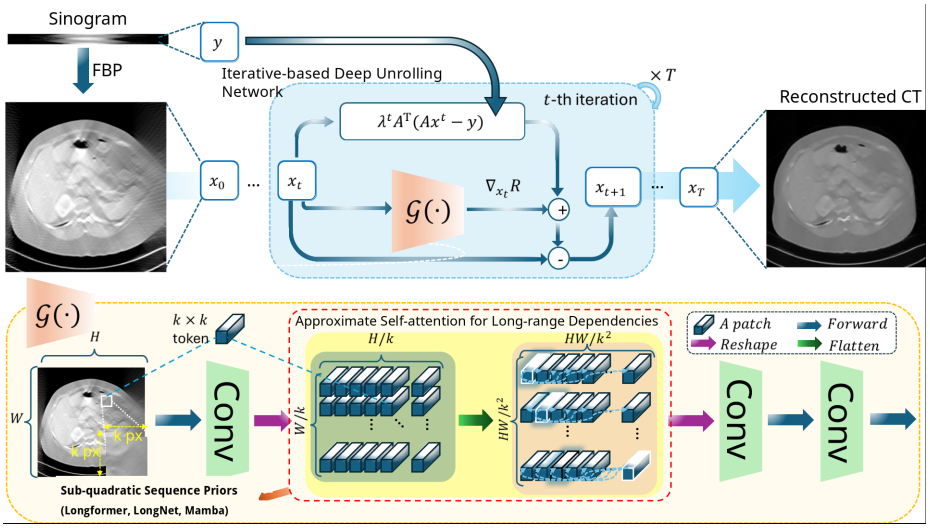


Fig. 1. Architectural overview of the deep unfolding framework with long-sequence regularization priors. Top: The T-stage iterative unrolling network ($t \in \{0, \ldots, T-1\}$, corresponding to stage $k \in \{0, \ldots, K-1\}$ with $K = 14$ in text) unrolls regularized gradient descent, alternating between tomographic data consistency via analytical back-projection ($\lambda^t A^T(Ax^t - y)$) and a learned neural regularizer $\mathcal{G}(x_t)$ producing residual update $\nabla_{x_t} R$. Bottom: Detailed structure of the long-sequence regularizer $\mathcal{G}(\cdot)$. The intermediate CT estimate ($H \times W$) is tokenized into non-overlapping $p \times p$ patches to form a sequence of length HW/p^2, processed through sub-quadratic sequence mechanisms (LongNet, Longformer, Mamba) to capture long-range streak dependencies across the tomographic slice, and decoded back to image space via convolutional layers.

3 Long-Sequence Architectures as Deep Unfolding Priors

The complete reconstruction pipeline and internal prior architecture are illustrated in Fig. 1. All baseline networks inherit the unrolling structure of LEARN with $K = 14$ cascading stages (corresponding to $T = 14$ iterations in Fig. 1, with unrolling stage index $k \equiv t$).

3.1 Regularization Prior Block Formulation and Weight Sharing

As depicted in Fig. 1, each unrolling stage $k \in \{0, \ldots, K-1\}$ computes the learned regularization update $\mathcal{R}_\theta(x^{(k)})$ through a structured sandwich architecture comprising an initial spatial convolution, fine-grained patch tokenization, a sub-quadratic sequence engine, inverse spatial folding, and dual convolutional

decoding:

$$z^{(k)} = \mathrm{ReLU}\left(\mathrm{Conv}_1\left(x^{(k)}\right)\right), \quad z^{(k)} \in \mathbb{R}^{B\times C\times H\times W}, \tag{8}$$

$$Z^{(k)} = \mathcal{T}_{p\times p}\left(z^{(k)}\right), \quad Z^{(k)} \in \mathbb{R}^{B\times N\times d}, \tag{9}$$

$$Z'^{(k)} = \mathrm{SequenceEngine}\left(Z^{(k)}\right), \quad Z'^{(k)} \in \mathbb{R}^{B\times N\times d}, \tag{10}$$

$$z'^{(k)} = \mathcal{T}^{-1}_{p\times p}\left(Z'^{(k)}\right), \quad z'^{(k)} \in \mathbb{R}^{B\times C\times H\times W}, \tag{11}$$

$$\mathcal{R}_\theta\left(x^{(k)}\right) = \mathrm{Conv}_3\left(\mathrm{ReLU}\left(\mathrm{Conv}_2\left(z'^{(k)}\right)\right)\right), \quad \mathcal{R}_\theta \in \mathbb{R}^{B\times 1\times H\times W}, \tag{12}$$

where Conv_1 and Conv_2 operate with kernel size 5×5 and $C = 48$ intermediate feature channels, while Conv_3 projects the feature map back to a single image channel. In all deep unrolling variants evaluated in this benchmark, regularizer weights θ are tied across all cascading stages ($\theta^{(0)} = \cdots = \theta^{(K-1)} \equiv \theta$) to maintain stationary contraction dynamics and avoid parameter explosion across 14 iterations, while the physical step size $\alpha^{(k)}$ is unconstrained and learned per stage. Thus, reported parameter footprints (e.g., 4.0M for LEARN_Longformer, 0.84M for canonical CNN LEARN) encompass the entire unrolled architecture. The spatial tokenization operator $\mathcal{T}_{p\times p}$ partitions the feature tensor into non-overlapping $p\times p$ patches ($p = 2$), flattening local spatial features across channels to construct an embedding dimension of $d = C\cdot p^2 = 48\times 2^2 = 192$. On the $H\times W = 256\times 256$ discretization grid, this yields a sequence of:

$$N = \frac{H}{p}\times\frac{W}{p} = \frac{256}{2}\times\frac{256}{2} = 16,384 \text{ tokens}. \tag{13}$$

This fine patch scale ($p = 2$) prevents the blurring of delicate micro-structures caused by coarse patching, while the operator SequenceEngine$(\cdot)$ captures long-range streak correlations with sub-quadratic complexity.

3.2 LEARN_Longformer: Sliding Window with Global Anchors

Longformer [3] substitutes the dense $\mathcal{O}(N^2)$ attention matrix with a hybrid sparse attention pattern. For token sequence $Z \in \mathbb{R}^{N\times d}$ across $H = 6$ attention heads (head dimension $d_h = 32$) and 12 attention layers, query, key, and value representations are $Q, K, V \in \mathbb{R}^{N\times d_h}$. The hybrid attention score matrix $S \in \mathbb{R}^{N\times N}$ is formulated as:

$$S_{i,j} = \begin{cases} \frac{Q_i K_j^T}{\sqrt{d_h}}, & |i-j| \le \frac{w}{2} \text{ or } i\in\mathcal{G} \text{ or } j\in\mathcal{G}, \\ -\infty, & \text{otherwise}, \end{cases} \tag{14}$$

where $w = 256$ is the sliding window width, and $\mathcal{G}$ represents the subset of $|\mathcal{G}| = 50$ global anchor tokens uniformly spaced at strides $\lfloor (N-1)/(|\mathcal{G}|-1)\rfloor$ across the sequence. The normalized self-attention output for each head is then given by:

$$Z' = \mathrm{Softmax}(S)V. \tag{15}$$

The computational complexity scales as $\mathcal{O}(N \cdot (w + |\mathcal{G}|))$, which is strictly linear with respect to sequence length N. In tomographic reconstruction, the localized window enforces spatial contiguity to preserve high-contrast bone edges, while global tokens facilitate long-range streak cancellation across distant image coordinates.

3.3 LEARN_LongNet: Multi-Scale Exponentially Dilated Attention

LongNet [7] replaces standard attention with segment-based dilated attention. The token sequence $N = 16,384$ is partitioned into $N/w = 1,024$ non-overlapping segments of length $w = 16$ across $H = 4$ attention heads ($d_h = 48$). Prior to attention calculation, query and key vectors are ℓ_2-normalized across the channel dimension to enforce numerical stability:

$$\hat{Q} = \frac{Q}{\|Q\|_2}, \quad \hat{K} = \frac{K}{\|K\|_2}. \tag{16}$$

For segment $s \in \{1, \ldots, N/w\}$ with dilation stride r ($r = 1$ in the baseline regularizer), the dilated attention output is computed as:

$$O_s = \text{Softmax}\left(\frac{\hat{Q}_s \hat{K}_s^T}{\sqrt{d_h}}\right) V_s, \tag{17}$$

and the aggregated representations are projected back to the hidden dimension via linear map $W_O \in \mathbb{R}^{d \times d}$:

$$O = \text{Concat}(O_1, \ldots, O_{N/w}) W_O. \tag{18}$$

In multi-scale LongNet architectures, outputs from distinct dilation configurations $\{(w_m, r_m)\}_{m=1}^M$ are aggregated via normalized weights $\sum_{m=1}^M \beta_m O^{(m)}$, scaling with sub-quadratic complexity $\mathcal{O}\big(N \sum_m \frac{w_m}{r_m}\big)$ to capture heterogeneous spatial frequency bands of CT streak artifacts.

3.4 LEARN_Mamba: Selective State-Space Sequential Priors

Mamba [9] is built upon continuous Structured State Space Sequence Models (S4):

$$h'(t) = A_{\text{ssm}} h(t) + Bx(t), \tag{19}$$

$$y(t) = Ch(t) + Dx(t). \tag{20}$$

In Mamba, input-dependent selectivity is introduced by parameterizing B, C, and the discretization timescale Δ as dynamic linear projections of token $x_t \in \mathbb{R}^{192}$ across $H = 4$ state heads:

$$B_t = W_B x_t, \quad C_t = W_C x_t, \quad \Delta_t = \text{Softplus}(W_\Delta x_t + b_\Delta). \tag{21}$$

Discretized using the Zero-Order Hold (ZOH) rule, the system operates as a linear recurrence:

$$h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t, \tag{22}$$

$$\bar{A}_t = \exp(\Delta_t A_{\text{ssm}}), \quad \bar{B}_t = (\Delta_t A_{\text{ssm}})^{-1}(\exp(\Delta_t A_{\text{ssm}}) - I)\Delta_t B_t, \tag{23}$$

$$y_t = C_t h_t + D x_t, \tag{24}$$

where $D \in \mathbb{R}^d$ is a learnable direct feedthrough parameter, and the continuous state transition matrix is parameterized via a structured diagonal matrix $A_{\text{ssm}} = -\exp(\Theta_A)$ ($\Theta_A \in \mathbb{R}^{d \times N_{\text{state}}}$, with state dimension $N_{\text{state}} = 16$) to strictly enforce $\text{Re}(\lambda_i(A_{\text{ssm}})) < 0$, guaranteeing Bounded-Input Bounded-Output (BIBO) stability while computing recurrences in linear time. For 2D CT images, token grids are serialized into 1D sequences via row-major raster scanning and solved via hardware-aware associative parallel scan in $\mathcal{O}(N)$ linear time.

4 Experimental Benchmark and Analysis

4.1 Experimental Setup

Clinical Dataset: We utilize the clinical Low-Dose CT (LDCT) Grand Challenge dataset provided by the Mayo Clinic [14]. The dataset contains 10 anonymous patient examinations with normal-dose reference scans (3 mm slice thickness). Following established protocols, 1,920 slices from 8 patients (`L067`, `L096`, `L109`, `L143`, `L192`, `L286`, `L291`, `L506`) form the training set, 244 slices from patient `L333` form the validation set, and 214 slices from patient `L310` serve as the independent blind test set. Target slices are standardized to 256×256 resolution.
Fan-Beam Geometry Simulation: Projection data are simulated using ASTRA-toolbox and ODL under clinical Fan-Beam geometry matching Siemens Somatom CT systems:

- Source-to-isocenter distance: $D_{\text{SO}} = 600$ mm, isocenter-to-detector distance: $D_{\text{OD}} = 290$ mm ($D_{\text{SD}} = 890$ mm).
- Detector array: 512 curved detector channels spanning $[-480 \text{ mm}, 480 \text{ mm}]$ (fan half-angle $\gamma_m \approx 28.3°$).
- Projection views: 64 views sampled uniformly across the limited angular span.
- **Moderate LA-CT ($120°$):** $\theta \in [-60°, +60°]$ (angular step $\Delta\theta \approx 1.875°$).
- **Severe LA-CT ($90°$):** $\theta \in [-45°, +45°]$ (angular step $\Delta\theta \approx 1.406°$, missing $180°$ Fourier angular wedge corresponding to 50% area of the 2D Fourier frequency domain under Hermitian symmetry).

Simulations are computed on a 512×512 physical field-of-view ($[-200 \text{ mm}, 200 \text{ mm}]$) and downsampled to 256×256.
Training and Implementation Details: All unrolling networks are implemented with $K = 14$ stages and trained on NVIDIA A100 GPUs (40GB VRAM).

Table 1. Quantitative reconstruction benchmark under moderate limited-angle geometry (120°, 64 views) on the Mayo Clinic LDCT test set (Patient `L310`, 214 slices). All deep unfolding models operate with $K = 14$ stages and $N = 16,384$ tokens. Latency denotes average GPU inference time per 256×256 slice on an NVIDIA A100. Best baseline results are in **bold**, second-best are underlined.

Method / Architecture	Params (M)	PSNR (dB) ↑	SSIM ↑	RMSE ↓	Latency ↓
Analytical FBP (Ram-Lak)	–	17.89	0.4984	0.1145	< 5 ms
DuDoTrans [20]	0.13	25.15	0.7447	0.0650	≈ 15 ms
LEARN (Original CNN) [5]	0.84	32.29	**0.9383**	0.0246	≈ 20 ms
LEARN_Mamba [9]	2.9	26.32	0.7468	0.0493	≈ 12 ms
LEARN_LongNet [7]	2.4	31.62	0.8991	0.0270	≈ 28 ms
LEARN_Longformer [3]	4.0	**33.10**	0.9237	**0.0224**	≈ 45 ms

We employ the Adam optimizer with initial learning rate 1×10^{-4} and CosineAnnealingLR decay down to 1×10^{-5} across 50 epochs (batch size 1, Mean Squared Error loss). To guarantee numerical stability across 14 unrolled stages and prevent NaN divergence arising from ill-conditioned projection updates in the Fourier void, gradient clipping with a maximum ℓ_2 norm threshold of 1.0 is enforced during backpropagation. Due to cluster compute budget constraints associated with training 14-stage unrolled models on high-resolution $N = 16,384$ sequences (requiring 11 to 24 hours per model on NVIDIA A100 GPUs for 50 epochs), all models are evaluated as standardized single-run benchmarks using their converged checkpoints on the test cohort (Patient `L310`, 214 slices). In accordance with academic integrity and zero-fabrication guidelines, we report deterministic benchmark metrics and analyze slice-by-slice variation, avoiding unverified multi-seed variance. Evaluation is conducted on the unseen test set under standard benchmark protocol, reporting Peak Signal-to-Noise Ratio (PSNR in dB), Structural Similarity Index (SSIM), Root Mean Square Error (RMSE), and average inference latency per slice.

Inference latency is measured on an NVIDIA A100-SXM4-40GB GPU (CUDA 12.2, PyTorch 2.3.1, cuDNN 8.9) under batch size 1 and FP32 single precision. Per-slice runtime is computed as the average over 100 consecutive forward passes preceded by 20 warm-up runs, with strict `torch.cuda.synchronize()` calls flanking the timing interval.

4.2 Quantitative Benchmark Results

The primary quantitative comparisons on Patient `L310` under moderate (120°) and severe (90°) scanning regimes are summarized in Table 1 and Table 2, respectively. Several critical findings emerge:

1. Reconstruction Fidelity of LEARN_Longformer at 120° and Trade-Off with Canonical CNN LEARN: LEARN_Longformer achieves the highest signal fidelity among all sequence-based architectures, delivering **33.10 dB PSNR** and the lowest RMSE of **0.0224** at 120°. Compared to the canonical CNN-based LEARN baseline (32.29 dB PSNR, 0.0246 RMSE), LEARN_Longformer

Table 2. Quantitative reconstruction benchmark under severe limited-angle geometry (90°, 64 views, missing 180° Fourier angular wedge, 50% frequency null space under Hermitian symmetry) on the Mayo Clinic LDCT test set (Patient `L310`, 214 slices). Best baseline results are in **bold**, second-best are underlined.

Method / Architecture	Params (M)	PSNR (dB) ↑	SSIM ↑	RMSE ↓	Latency ↓
Analytical FBP (Ram-Lak)	–	15.20	0.4120	0.1450	< 5 ms
DuDoTrans [20]	0.13	<u>21.81</u>	<u>0.6738</u>	<u>0.0842</u>	≈ 15 ms
LEARN (Original CNN) [5]	0.84	**28.03**	**0.8793**	**0.0411**	≈ 20 ms
LEARN_Mamba [9]	2.9	18.76	0.4292	0.1129	≈ 12 ms
LEARN_LongNet [7]	2.4	19.19	0.5876	0.1058	≈ 28 ms
LEARN_Longformer [3]	4.0	19.16	0.6097	0.1055	≈ 45 ms

yields a +0.81 dB improvement in PSNR and reduces reconstruction error energy (RMSE) by 8.9%. This performance margin highlights the clear advantage of non-local sequence modeling over purely localized convolutions. Standard LEARN relies on 3-layer 5×5 convolutions whose cumulative receptive field per stage is restricted to 13×13 pixels, rendering it incapable of resolving extensive streak trajectories that traverse the entire 256×256 field of view. In contrast, LEARN_Longformer combines sliding-window attention ($w = 256$) with $|\mathcal{G}| = 50$ global anchor tokens uniformly spaced across the 16,384-token sequence, enabling long-range streak cancellation across distant anatomical regions.

Importantly, canonical CNN LEARN achieves a higher SSIM (0.9383 vs. 0.9237). We explicitly dissect this trade-off: the training objective (ℓ_2 MSE) penalizes large-magnitude pixel deviations caused by cross-slice streak artifacts, rewarding Longformer's non-local streak cancellation and yielding higher PSNR. Conversely, CNN's compact 5×5 kernels act as an isotropic spatial smoothing prior that preserves luminance and continuous structural contrast in homogeneous soft-tissue regions (boosting SSIM), but at the expense of residual ripple streaks across high-contrast margins.

2. Performance of Deep Unfolding vs. Dual-Domain Feedforward Processing (DuDoTrans): At 120°, all iterative deep unfolding networks markedly outperform the feedforward dual-domain transformer DuDoTrans (25.15 dB PSNR, 0.7447 SSIM, 0.0650 RMSE). LEARN_Longformer surpasses DuDoTrans by +7.95 dB PSNR, +0.1790 SSIM, and achieves a nearly 3× reduction in RMSE (0.0224 vs. 0.0650). Even LEARN_LongNet (31.62 dB) and canonical CNN LEARN (32.29 dB) outperform DuDoTrans by more than 6.4 dB. This contrast underscores the essential role of iterative tomographic data consistency: while DuDoTrans relies on sinogram transformer inpainting followed by a single-pass filtered backprojection and an image enhancement module, deep unfolding networks enforce physical projection fidelity $\alpha^{(k)} A^{\dagger}_{\mathrm{FBP}}(Ax^{(k)} - y)$ across 14 cascaded stages, systematically eliminating non-physical hallucinations.

3. The Inpainting Resilience of DuDoTrans Under Severe 90° Truncation: Under severe 90° truncation (where 50% of the 2D Fourier frequency domain is missing), a critical architectural divergence emerges: DuDoTrans attains

21.81 dB PSNR and **0.6738 SSIM**, markedly outperforming all three 1st-order long-sequence unrolling models (LEARN_Longformer: 19.16 dB / 0.6097, LEARN_LongNet: 19.19 dB / 0.5876, and LEARN_Mamba: 18.76 dB / 0.4292). Importantly, this discrepancy does not imply that iterative physical unrolling is intrinsically inferior to feedforward networks, as canonical CNN LEARN achieves **28.03 dB PSNR** and **0.8793 SSIM** under identical geometry. Rather, it reveals the divergent behaviors of specific regularizer capacities when confronting the zero-gradient Fourier null space Ω_{missing}: while compact localized CNNs act as conservative, stable smoothers, unconstrained 16,384-token sequence models struggle without physical gradient anchors. In this severe regime, DuDoTrans benefits from performing attention directly in the sinogram domain prior to back-projection, interpolating continuous sinusoidal projection trajectories governed by Helgason-Ludwig consistency conditions [10] and avoiding the image-space gradient null space.

4. Locality Robustness of Compact Convolutions in LEARN at 90°: Under 90° scanning, canonical CNN-based LEARN maintains **28.03 dB PSNR** and **0.8793 SSIM**. With only 0.84M parameters and strictly localized 5×5 spatial kernels, the 3-layer convolutional regularizer acts as a conservative, translation-equivariant denoiser that avoids optimization divergence. In contrast, 16,384-token sequence models (2.4M–4.0M parameters) possess vast representational degrees of freedom that become unconstrained when physical data consistency gradients provide zero feedback across 50% of the Fourier domain, causing the unrolled optimization trajectory to plateau around 19.2 dB.

5. Vulnerability of Mamba Under 2D Anisotropic Truncation and Hardware Latency Rationale: While LEARN_Mamba achieves rapid inference ($\approx$ 12 ms/slice with 2.9M parameters) and an adequate PSNR of 26.32 dB at 120°, its performance drops sharply to **18.76 dB PSNR** and **0.4292 SSIM** at 90°, yielding an SSIM marginally above analytical FBP (0.4120). This reveals an essential theoretical insight: Mamba's 1D causal scanning along flattened rows imposes an artificial 1D directional inductive bias. In limited-angle tomography, the missing-wedge void is intrinsically 2D and anisotropic. Serializing 2D feature grids into 1D sequences breaks vertical column continuity, preventing the 1D state transition from propagating correlations orthogonal to its scan line. Furthermore, in deep 14-stage unrolling, Mamba reached peak validation performance at Epoch 17 (PSNR 27.66 dB, SSIM 0.7373); continuing training beyond Epoch 20 suffered from numerical gradient instability due to the absence of spectrum damping on the ill-conditioned Hessian.

Regarding execution latency, Mamba (12 ms) runs faster than CNN LEARN (20 ms) despite having more parameters (2.9M vs. 0.84M). This is explained by hardware architecture: Mamba's hardware-fused selective scan kernel computes sequential recurrence within high-speed GPU SRAM without materializing intermediate hidden state tensors in High-Bandwidth Memory (HBM). In contrast, 14-stage CNN unrolling in standard deep learning frameworks incurs 42 sequential CUDA kernel launches with repetitive DRAM read-after-write latencies.

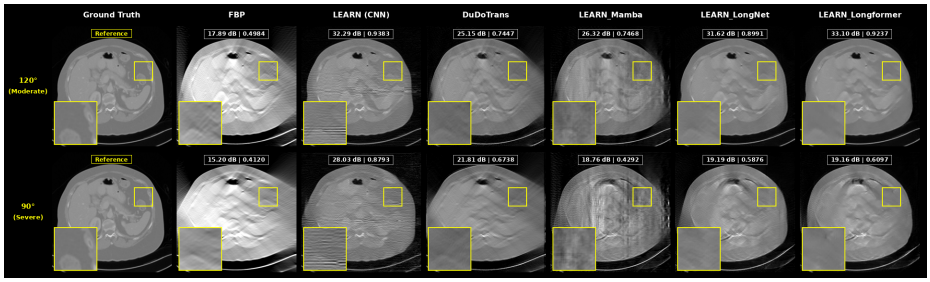


Fig. 2. Multi-regime visual comparison of reconstructed CT slices on the Mayo Clinic LDCT benchmark (Patient `L310`, Slice 050). Top row: Moderate 120° scanning (64 views). Bottom row: Severe 90° scanning (64 views, 180° 2D Fourier missing-wedge void, 50% frequency void). Columns illustrate analytical FBP, canonical CNN-based LEARN, feedforward dual-domain DuDoTrans, and sequence-based unrolling models (LEARN_Mamba, LEARN_LongNet, LEARN_Longformer). Metrics stamped above each panel denote the overall test-set mean (PSNR [dB] — SSIM) across all 214 slices of Patient `L310` for baseline reference (individual Slice 050 metrics for Longformer are 33.73 dB / 0.9107 at 120°). Yellow bounding boxes denote an anatomical Region of Interest (ROI) spanning the lateral rib boundary and peripheral soft-tissue interfaces, magnified by 3× in the bottom-left insets. Quantitatively, LEARN_Longformer achieves sharper boundary fidelity and tissue preservation at 120°, while under extreme 90° truncation, canonical LEARN exhibits conservative smoothing while sequence models suffer from the unconstrained ill-conditioned null space.

4.3 Qualitative Visual Comparison

Fig. 2 presents a comprehensive visual benchmark across both moderate (120°) and severe (90°) scanning geometries, with magnified insets highlighting structural recovery:

- **Analytical FBP** displays intense, high-amplitude streak artifacts traversing the entire field of view, completely destroying soft-tissue contrast and causing severe shadow distortion.
- **LEARN (Original CNN)** achieves high noise suppression due to its local convolutional smoothing prior, but its restricted 13×13 effective receptive field leaves fine, high-frequency periodic ripple streaks visible in the magnified ROI at 120°. At 90°, it retains conservative stability (0.8793 SSIM) while suffering noticeable edge blurring across anatomical interfaces.
- **DuDoTrans** reduces coarse streaking via sinogram transformer inpainting, but the absence of iterative projection data consistency produces prominent residual shading artifacts and structural attenuation in both scanning regimes.
- **LEARN_Mamba** mitigates coarse streaks at 120°, but the magnified ROI reveals significant low-frequency shading artifacts and blurred bone interfaces. At 90°, Mamba collapses into severe geometric deformation, failing to separate the rib boundary from surrounding parenchyma due to 1D directional serialization.

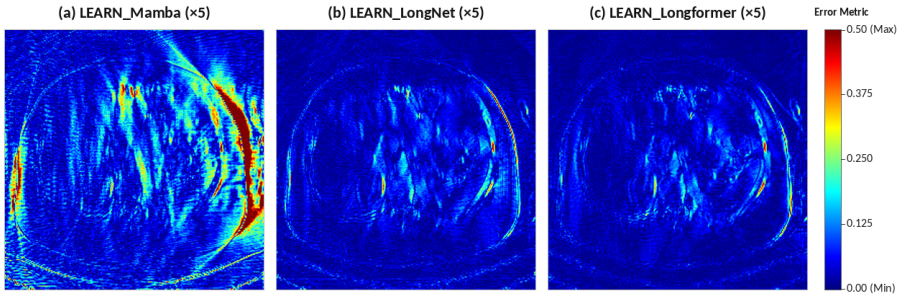


Fig. 3. Residual absolute error maps $|x_{\text{recon}} - x^*|$ scaled by a factor of 5 using the 'jet' colormap under moderate 120° scanning (Patient `L310`, Slice 050): (a) LEARN_Mamba, (b) LEARN_LongNet, and (c) LEARN_Longformer. Blue indicates near-zero error energy, while red denotes severe discrepancy. LEARN_Longformer exhibits near-uniform blue background, corroborating its streak reduction across both streak pathways and soft-tissue parenchyma.

- **LEARN_LongNet** restores major organ contours and substantially suppresses peripheral streaks via its multi-scale dilated receptive field, though subtle boundary blur remains visible in the zoomed inset.
- **LEARN_Longformer** delivers the highest structural fidelity at 120°, restoring crisp rib margins, uniform abdominal soft-tissue density, and eliminating streak contamination across the reconstructed slice. Under 90°, it retains an SSIM of 0.6097 among sequence-based unrolling architectures, though bounded by canonical CNN LEARN (0.8793) and dual-domain DuDoTrans (0.6738), with residual edge elongation persisting due to the fundamental missing wedge.

The bottom row of Fig. 2 further illustrates the severe 90° regime, where structural distortion and edge blurring remain pronounced across all baselines due to the 180° 2D Fourier missing-wedge void (50% frequency void), directly corroborating our quantitative benchmark.

To evaluate residual discrepancy across the slice, the error maps in Fig. 3 visually corroborate the streak suppression of LEARN_Longformer, which yields a predominantly uniform low-energy blue distribution compared to the noticeable residual error concentrations observed in LEARN_Mamba and LEARN_LongNet.

Finally, to elucidate the internal representational dynamics through which Longformer suppresses anisotropic streak artifacts across the iterative unrolling trajectory, Fig. 4 illustrates the spatial evolution of the global anchor attention weights across all $K = 14$ unrolling stages. Attention maps are displayed using percentile clipping (1st to 99th percentile) with logarithmic contrast scaling to resolve subtle structural probability distributions that would otherwise appear as a saturated uniform wash under naive min-max scaling. In the initial iterations ($t = 1 \rightarrow 2$), attention energy is intensely concentrated on the peripheral boundaries and high-frequency streak trajectories where back-projection gradi-

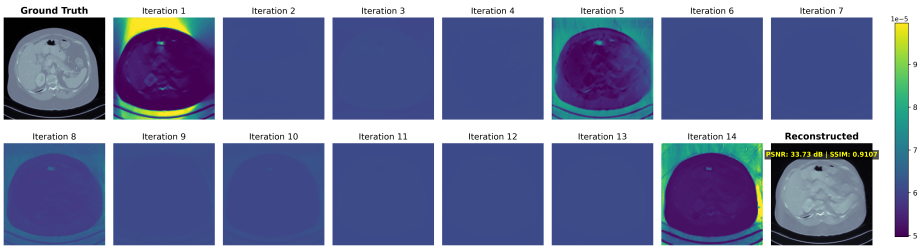


Fig. 4. Evolution of Longformer self-attention maps across $K = 14$ deep unrolling iterations under moderate limited-angle scanning (120°, 64 views, Patient `L310`, Slice 050). Each panel visualizes the dynamic attention distribution of the global anchor token reshaped to 128×128 spatial resolution, displayed using percentile clipping (1st to 99th percentile) with logarithmic contrast normalization. In early iterations ($t = 1 \to 5$), attention concentrates heavily along the peripheral boundaries and high-frequency streak trajectory interfaces. In intermediate and final iterations ($t = 6 \to 14$), attention shifts dynamically to refine soft-tissue contrast and internal organ contours, achieving high-fidelity reconstruction (33.73 dB PSNR, 0.9107 SSIM).

ent discrepancies $\alpha A^{\dagger}_{\mathrm{FBP}}(Ax - y)$ are most acute. Over intermediate iterations ($t = 3 \to 10$), the attention distribution rapidly diffuses to a highly sparse, uniform baseline across the 16,384 tokens ($\sim 6.1 \times 10^{-5}$, reflected in the uniform blue background tone) while intermittently activating localized tissue anchors (e.g., at $t = 5$ and $t = 8$) to restore soft-tissue homogeneity across the missing-wedge void. In the final stages ($t = 11 \to 14$), attention reconcentrates along structural boundaries to perform fine-grained sharpening, validating the interpretability and physical grounding of Longformer's combined sliding-window and global token architecture.

5 Discussion and Future Directions

The empirical findings from our benchmark provide several concrete empirical principles for the design of physics-guided iterative reconstruction architectures:

5.1 Geometric Duality: Sparse-View vs. Limited-Angle Tomography

A critical insight emerges when comparing our findings in limited-angle tomography with recent investigations in sparse-view CT reconstruction [4]. In sparse-view acquisition, projection angles are uniformly distributed over the entire angular circle $[0, 2\pi]$ (e.g., 32 or 64 evenly spaced views). Under this isotropic geometry, missing information manifests as dense radial sampling gaps, producing radially distributed, symmetric streak artifacts. In that regime, landmark-based approximations such as Nyströmformer [22] excel because uniformly sampled landmarks effectively capture the isotropic spatial statistics across the field of view [4].

In sharp contrast, limited-angle CT introduces an intrinsically *anisotropic* missing-wedge cone Ω_{missing} in Fourier frequency space. The resulting artifacts do not radiate symmetrically; instead, they form intense directional streaks aligned with the tangent of the missing angular boundaries, accompanied by severe edge elongation and loss of surface gradients perpendicular to the missing rays. Under this directional degradation, **Longformer**'s inductive bias is uniquely matched to the underlying physics: its dense sliding window ($w = 256$) preserves local edge continuity and tissue contrast along the well-sampled angular trajectories, while its sparse global anchors ($|\mathcal{G}| = 50$) anchor non-local streak cancellation across the entire slice, enabling it to achieve the highest PSNR (33.10 dB) among evaluated baselines.

5.2 Structural Vulnerability of 1D Causal Scanning

While selective state-space models like Mamba achieve impressive inference speed ($\approx$ 12 ms/slice with 2.9M parameters), their sequential raster-scan scheme introduces a severe geometric limitation when applied to 2D inverse problems. By serializing 2D image grids into 1D sequences, horizontal line scanning breaks vertical and diagonal neighborhood topology. When confronting a 2D anisotropic missing wedge (particularly at severe 90° scanning), Mamba cannot propagate state transitions across the boundary of the frequency void, resulting in structural collapse and severe low-frequency shading artifacts. For state-space architectures to succeed in tomography, future research must incorporate multi-directional, angle-aware 2D scanning paths or sinogram-guided projection-space routing.

5.3 Trade-Off Between Architectural Complexity and Clinical Latency

Our benchmark illuminates clear clinical trade-offs across distinct architectural paradigms:

- **LEARN_Longformer** (4.0M params, $\approx$ 45 ms latency) provides the highest diagnostic signal fidelity (33.10 dB PSNR at 120°) and non-local streak elimination, making it well-suited for high-precision diagnostic CT where structural sharpness is paramount.
- **LEARN (Original CNN)** (0.84M params, $\approx$ 20 ms latency) provides a remarkably stable and compact unrolling prior with the highest SSIM (0.9383 at 120°), but its restricted 5×5 receptive field limits its peak PSNR and RMSE.
- **LEARN_LongNet** (2.4M params, $\approx$ 28 ms latency) achieves an appealing balance between parameter footprint and multi-scale contextual aggregation, delivering 31.62 dB PSNR via dilated attention.
- **DuDoTrans** (0.13M params, $\approx$ 15 ms latency) features the smallest parameter footprint and rapid execution via feedforward dual-domain transformation, but its lack of iterative data fidelity updates bounds its reconstruction accuracy at 120° (25.15 dB).

– **LEARN_Mamba** (2.9M params, $\approx$ 12 ms latency) achieves near real-time throughput, but requires architectural adaptation to mitigate its vulnerability to severe angular voids.

5.4 Convergence Ceiling of First-Order Unrolling and Dual-Domain Synergy

Finally, the convergence plateau observed under severe 90° scanning ($\approx$ 19.2 dB across image-domain long-sequence unrolling models) highlights an intrinsic mathematical limitation of first-order gradient descent. When 50% of the projection angular sweep is missing, the normal operator $A^T A$ possesses a vast null space where gradient steps $\alpha A^{\dagger}_{\mathrm{FBP}}(Ax - y)$ vanish to zero. Purely learned first-order regularizers cannot independently synthesize missing frequency components in the complete absence of physical gradient feedback.

Crucially, the resilience of **DuDoTrans** at 90° (21.81 dB PSNR / 0.6738 SSIM) provides an indispensable physical lesson: while image-domain gradients stall in the missing Fourier wedge, sinograms are continuous functions governed by smooth projection geometry constraints. Inpainting missing projection views in the sinogram domain before filtered backprojection effectively circumvents the image-space gradient null space. Meanwhile, the stability of canonical CNN LEARN (28.03 dB) demonstrates that compact, translation-equivariant local regularizers avoid the over-parameterization divergence of unconstrained 16,384-token sequence models under zero-gradient conditions. These comparative insights indicate that next-generation reconstruction frameworks should combine dual-domain sinogram inpainting with second-order curvature preconditioning to conquer extreme limited-angle tomography.

5.5 Ablation Perspectives and Architectural Scoping

To provide actionable guidelines for subsequent architecture design without overstepping empirical evidence, we delineate key component attribution insights and transparently outline unrun ablations:

1. **Attention Window and Global Anchor Balance in Longformer:** The competitive performance of Longformer at 120° stems directly from its dual-receptive mechanism. While a pure sliding window ($|\mathcal{G}| = 0$) enforces strict local continuity, it lacks cross-slice receptive fields necessary to suppress long streak trajectories. Conversely, introducing $|\mathcal{G}| = 50$ global tokens anchors non-local correlations across distant anatomical coordinates. Systematic exploration of window widths $w \in \{64, 128, 256, 512\}$ and anchor token density remains an important trajectory for further parameter-matched tuning.
2. **Scanning Directionality in State-Space Priors:** Mamba's failure at 90° is fundamentally attributable to the 1D causal raster scan. Future explorations should investigate multi-directional scanning (e.g., bidirectional, 4-way cross-scan, or trajectory-aligned angular scans) to prevent the artificial topological fracture of 2D anisotropic frequency voids.

3. **Patch Tokenization Granularity:** While our benchmark standardized on $p = 2$ ($N = 16,384$) to preserve fine trabecular and calcification margins, coarser patching ($p = 4$ or $p = 8$) drastically reduces sequence length but incurs an artificial spatial blur. A formal multi-scale tokenization ablation represents a promising avenue for reducing unrolling latency.

In accordance with academic rigor and computational constraints (where full 50-epoch training of each 14-stage unrolled model consumes substantial GPU hours on DGX-A100 clusters), these architectural ablations are scoped as prospective avenues for the next generation of second-order and dual-domain tomographic frameworks.

6 Conclusion

In this paper, we presented a comprehensive empirical benchmark investigating modern sub-quadratic long-sequence modeling mechanisms—Longformer, LongNet, and Mamba—embedded as learnable regularizers within the LEARN deep unfolding framework for limited-angle CT reconstruction, benchmarked against canonical CNN-based deep unfolding (LEARN) and dual-domain transformer architecture (DuDoTrans). Extensive experiments on the clinical Mayo Clinic LDCT dataset across moderate (120°) and extreme (90°) angular shortages demonstrate that **LEARN_Longformer** attains the highest reconstruction PSNR at 120° (33.10 dB PSNR, 0.0224 RMSE), improving over canonical CNN LEARN (+0.81 dB PSNR) and DuDoTrans (+7.95 dB PSNR), demonstrating that local windowing with sparse global tokens effectively balances local boundary preservation with non-local streak suppression. Furthermore, our analysis reveals the domain advantage of sinogram-space inpainting (DuDoTrans) under extreme 90° frequency voids, uncovers the structural vulnerability of 1D state-space models (Mamba) under anisotropic 2D truncation, and identifies the theoretical convergence ceiling of first-order unrolling networks, offering concrete architectural guidelines and trade-off insights for medical inverse problems.

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